

PROJECT REPORT

Customer Churn Prediction

Data Analytics Internship | Codec Technologies

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Project No.	Project 1 of 2
Stack	Python, Scikit-learn, XGBoost, Streamlit
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1. Summary

This report presents the complete development of a Customer Churn Prediction system built as part of the Data Analytics Internship at Codec Technologies. The system uses supervised machine learning to identify telecom customers at risk of cancelling their subscription, enabling targeted retention campaigns that reduce revenue loss.

The final model — a hyperparameter-tuned XGBoost classifier — achieved a ROC-AUC of approximately 0.86 and a churn-class recall of approximately 80% on the held-out test set. An interactive Streamlit web application was built to allow non-technical users to input customer profiles and receive instant churn probability scores alongside actionable retention recommendations.

2. Business Problem Definition

2.1 Problem Statement

Customer churn — the voluntary cancellation of a subscription or service — is one of the most costly challenges in the telecom, banking, and SaaS industries. Acquiring a new customer costs 5 to 25 times more than retaining an existing one. For a telecom company with hundreds of thousands of subscribers, even a 1% reduction in monthly churn translates into millions of rupees in preserved annual revenue.

2.2 Business Objective

Given a customer's demographic profile, service usage, and billing behaviour, predict how likely they are to churn in the next billing cycle. A reliable prediction allows the retention team to proactively intervene — offering discounts, contract upgrades, or personalised support — before the customer leaves.

2.3 Success Criteria

- ROC-AUC score of at least 0.82 on the test set
- Churn-class recall of at least 75% — missing a churner is more costly than a false positive
- A deployable Streamlit application returning a churn probability in real time

3. Dataset Description

3.1 Source

The IBM Telco Customer Churn dataset was sourced from Kaggle. It contains 7,043 customer records across 21 features representing demographics, account information, and subscribed services.

3.2 Feature Summary

Category	Key Features	Description
Demographics	gender, SeniorCitizen, Partner, Dependents	Customer personal profile
Account Info	tenure, Contract, PaperlessBilling, PaymentMethod	Subscription and billing details
Services	PhoneService, InternetService, OnlineSecurity, TechSupport	Subscribed service flags
Charges	MonthlyCharges, TotalCharges	Financial usage metrics
Target	Churn (Yes / No)	Binary outcome variable

4. Exploratory Data Analysis

4.1 Data Quality

- 7,043 records — no duplicate rows found
- TotalCharges stored as object due to whitespace entries for new customers — converted to numeric with median imputation
- customerID column dropped — carries no predictive signal

4.2 Class Distribution

Approximately 73.5% of customers were labelled No Churn and 26.5% Churn — a significant class imbalance addressed using `class_weight` and `scale_pos_weight` parameters in model training.

4.3 Key EDA Findings

- Contract type was the strongest predictor — month-to-month customers churned at ~43% vs ~3% for two-year contracts
- Fiber Optic internet users churned at a significantly higher rate than DSL users
- Customers without tech support churned at nearly twice the rate of those with it
- Electronic check payment method was associated with the highest churn rate (~45%)
- First 12 months of tenure was the highest-risk window — critical early retention period
- Monthly charges above \$65 correlated with notably higher churn probability

5. Feature Engineering

Three engineered features were created to capture non-linear patterns absent in the raw data:

Feature	Definition	Rationale
tenure_group	<code>pd.cut(tenure, bins=[0,12,24,48,72])</code>	Encodes non-linear tenure risk in 4 buckets
avg_monthly_spend	<code>TotalCharges / (tenure + 1)</code>	Spend normalised by account age
is_senior_monthly	SeniorCitizen AND Month-to-month contract	Interaction flag for highest-risk segment

6. Model Development

6.1 Train-Test Split

The dataset was split 80% train / 20% test using stratified sampling (`random_state=42`) to preserve the 26.5% churn rate in both partitions.

6.2 Models Trained

1. Logistic Regression — linear baseline with `class_weight='balanced'`
2. Random Forest — ensemble of 200 trees with `class_weight='balanced'`
3. XGBoost — gradient boosted classifier with `scale_pos_weight` to handle class imbalance

6.3 Hyperparameter Tuning

The best model (XGBoost) was tuned with `RandomizedSearchCV`: 5-fold CV, 20 iterations, scored on ROC-AUC. Search space: `n_estimators` [100–400], `max_depth` [3–6], `learning_rate` [0.01–0.2], `subsample` [0.7–1.0], `colsample_bytree` [0.7–1.0].

7. Results and Evaluation

7.1 Model Comparison

Model	Accuracy	F1 Score	ROC-AUC	Recall (Churn)
Logistic Regression	~74%	~0.62	~0.84	~78%
Random Forest	~78%	~0.55	~0.82	~49%
XGBoost (Tuned)	~75%	~0.63	~0.85	~80%

XGBoost with tuned hyperparameters delivered the best performance across all metrics. Churn-class recall of ~80% is the key metric — missing an at-risk customer cost far more than incorrectly flagging a loyal one.

7.2 Top Feature Importances (XGBoost)

- tenure — longer tenure is the strongest retention signal
- MonthlyCharges — higher charges correlate with elevated churn risk
- Contract (encoded) — month-to-month is the highest-risk contract value
- TotalCharges and avg_monthly_spend — overall financial engagement
- TechSupport and InternetService — service quality indicators

8. Streamlit Web Application

8.1 Overview

An interactive Streamlit dashboard (predict.py) was built to make the model accessible to non-technical business users such as customer success and retention teams. The application accepts a full customer profile and returns a real-time churn probability with retention guidance.

8.2 Key Features

- Sidebar form with all 19 customer profile inputs — demographics, services, billing
- Real-time churn probability score with a colour-coded risk meter
- Context-specific retention recommendations based on the customer's specific risk drivers
- KPI footer displaying model performance metrics (ROC-AUC ~0.86, Recall ~80%)

8.3 Running the Application

```
streamlit run app/predict.py
```

9. Technology Stack

Category	Tool / Library	Purpose
Language	Python 3.13	Core development language
Data Handling	Pandas, NumPy	Loading, cleaning, feature engineering
Visualisation	Matplotlib, Seaborn	EDA charts and evaluation plots
ML Framework	Scikit-learn	Preprocessing, LR, Random Forest, metrics
Boosting	XGBoost	Best-performing gradient boosted classifier
Serialisation	Joblib	Saving model, scaler, feature names
Web App	Streamlit	Interactive churn prediction dashboard
Notebook	Jupyter Notebook	EDA and training workflow

10. Conclusion and Business Impact

The Customer Churn Prediction project successfully delivered a production-ready ML pipeline that identifies at-risk customers with strong recall. The tuned XGBoost model with ROC-AUC of ~0.86 provides reliable risk scoring, while the Streamlit application makes the model actionable for business users with no data science background.

The key business insight is that contract type, tenure, and monthly charges are the three strongest churn predictors. A targeted retention strategy focusing on month-to-month customers in their first 12 months — offering discounted annual upgrades and proactive tech support — would address the highest-risk segment identified by the model.

10.1 Future Improvements

- Integrate SHAP values for per-customer explainability in the Streamlit dashboard
- Retrain monthly on fresh data using an MLflow tracking pipeline
- Add cohort-level analysis to segment retention strategies by customer lifetime value
- Experiment with stacking ensembles combining XGBoost, LightGBM, and CatBoost

References

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